Assignment 1: MECHENG 4 AI 3

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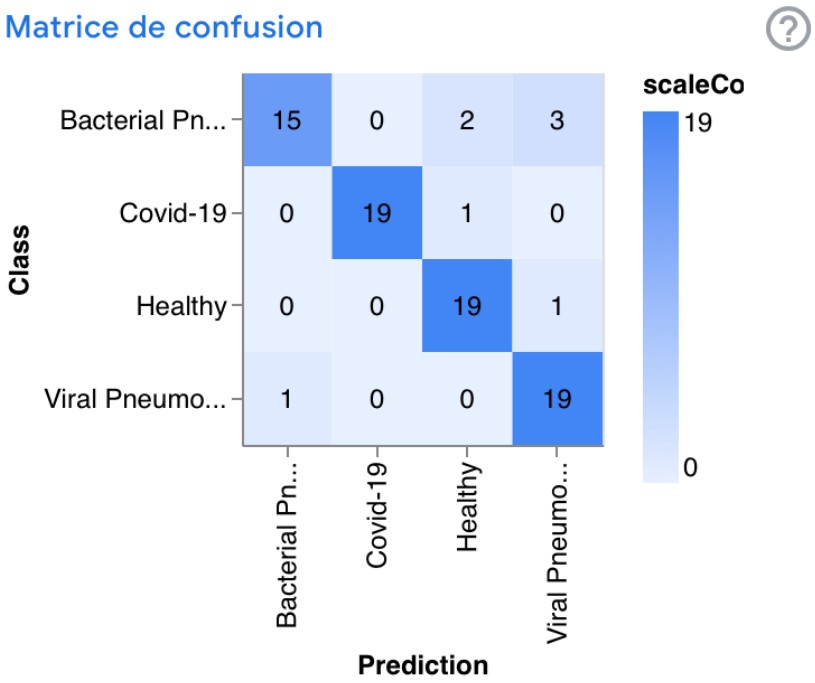
1. Model Training

We trained a deep neural network model using Google Teachable Machine on the provided dataset, which consists of 133 chest X-ray images. These images are divided into four classes:

1. Health
2. COVID-19
3. Bacterial Pneumonia
4. Viral Pneumonia

The model was trained using the following hyperparameters:
Epochs = 50, Batch size = 16, Learning rate = 0.001

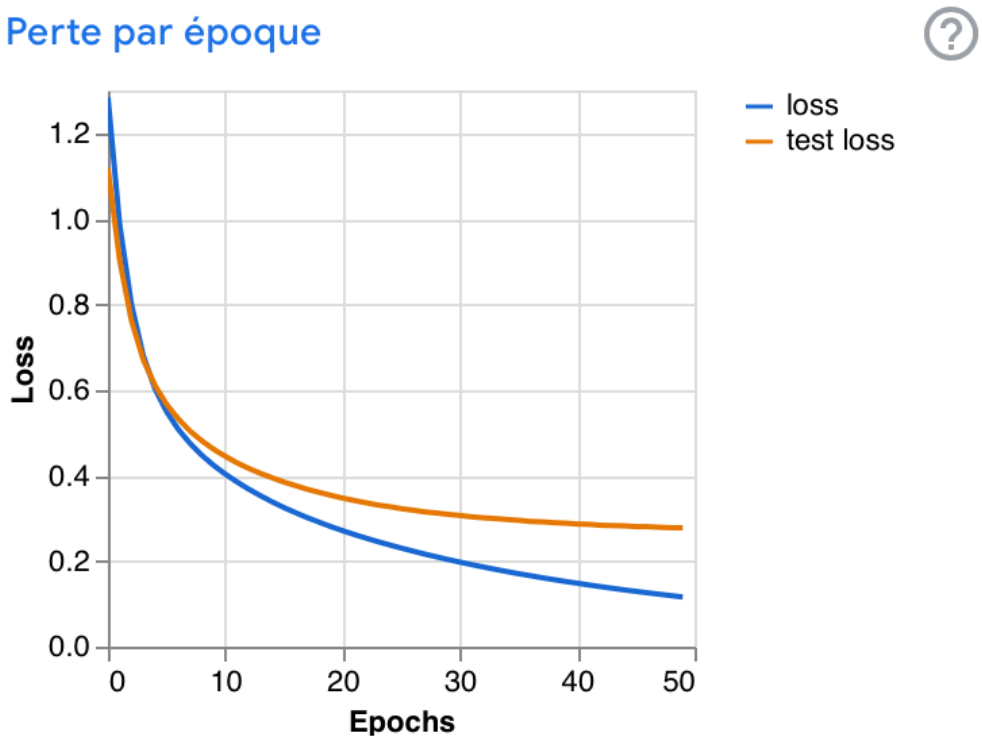
2. Model performance



Confusion Matrix

The confusion matrix shows that the model performs well across most classes. COVID-19, Healthy, and Viral Pneumonia are classified correctly in 19 out of 20 cases. However, Bacterial Pneumonia is confused 2 times with Healthy and 3 times with Viral Pneumonia. This indicates that Bacterial Pneumonia is the most challenging class for the model to distinguish.

3. Loss Curve Analysis



Loss Curve

Both the training and testing loss curves decrease gradually and converge, which indicates stable learning. Toward the end of training, there is a small divergence between the two curves of about 0.2 loss units. However, this gap is not large enough to clearly indicate overfitting or underfitting. Overall, the model shows good convergence behavior.

5. Hyperparameter Tuning

We tested the above combination of the different hyperparameters:

1) Epoch = 100, batch size = 32, learning rate = 0.001
   The model performed very well on the training data (accuracy = 1) with a good accuracy but less on the test data (accuracy = 0.87) suggesting a slight overfitting model. Also it took more time to train du to the increasing number of epochs.

2) Epoch = 50, batch size = 16, learning rate = 0.00001
   Due to the low learning rate, the model didn't converge and it's less accurate compared to the first model (Test accuracy = 0.80). However the loss curve on the train and test data are similar and the classification time increases.

3) Epoch = 50, batch size = 16, learning rate = 0.0005
   The model performed well with a test accuracy of 0.93, the test loss decreased gradually with the raining loss and it took less time than the previous two models.

Conclusion

Overall, the third hyperparameter combination provides the best trade-off between accuracy, convergence, and training time. The dataset is relatively small (133 images), which increases the risk of overfitting. With more data and further tuning, this model could be improved and made more suitable for clinical use.